7.5.3 Autonomous weapons systems and the ethics of drone warfare

The emergence of autonomous weapon systems (AWS) drones capable of identifying, targeting, and engaging adversaries without direct human control represents one of the most profound ethical challenges in modern warfare. These systems mark a paradigm shift from remotely piloted drones, where human operators retain decision-making authority, to machines that can make lethal decisions independently using artificial intelligence (AI), machine learning, and sensor fusion. While such technologies promise faster response times, reduced human casualties, and improved precision in combat, they also raise deep concerns about accountability, moral responsibility, and compliance with international humanitarian law (IHL).

A key ethical dilemma lies in the delegation of life-and-death decisions to algorithms. Human operators are traditionally bound by moral reasoning, empathy, and legal frameworks such as the Geneva Conventions, which emphasize proportionality, distinction, and necessity in the use of force. In contrast, autonomous systems operate based on programed logic and data patterns, devoid of emotional or ethical context. This raises the question: can a machine truly make morally justifiable decisions in war? The possibility of unintended civilian casualties due to misidentification, system malfunction, or data bias highlights the grave moral risks of removing human oversight from combat operations.

The issue of accountability further complicates the ethical landscape. When an autonomous drone commits an unlawful act, such as targeting non-combatants, determining liability becomes ambiguous. Should responsibility rest with the programer, manufacturer, military commander, or the state that deployed the system? This “responsibility gap” undermines